David Mohaisen, Professor
University of Central Florida
Department of Computer Science
Cybersecurity and Privacy Cluster

https://cs.ucf.edu/~mohaisen/

Coreference Resolution and Entity Tracking

① Introduction
② Technical details
③ Challenges
④ Examples
⑤ More examples
⑥ Summary and Takeaways

Objectives

- Understand **coreference resolution** and its role in text understanding.
- Distinguish **coreference vs. anaphora**.
- Compare modeling approaches: **mention pair, mention ranking, entity-level, end-to-end neural**.
- Analyze challenges in **mention detection and spurious mentions**.
- Understand the role of **knowledge-based and neural contextual models**.

Coreference Resolution

Coreference Resolution

- **What is Coreference Resolution?**
  - The task of **identifying all mentions** in a text that refer to the **same real-world entity**.
  - Essential for **text understanding, dialogue systems, and machine translation**.
- **Example:** *"Barack Obama nominated Hillary Rodham Clinton as his secretary of state on Monday. He chose her because she had foreign affairs experience as a former first lady."*
- **Entities & Coreferences:**
  - *"Barack Obama"* → "**He**"
  - *"Hillary Rodham Clinton"* → "**her**", "**she**", "**former first lady**"

- **Why is Coreference Resolution Important?**
  - **Improves text coherence** for summarization and dialogue systems.
  - **Enhances machine comprehension** in tasks like question answering.
  - **Aids machine translation** by resolving ambiguous pronoun references.

Coreference Resolution: Identifying Mentions

- Identify all mentions that refer to the same real world entity
  - **Barack Obama** nominated **Hillary Rodham Clinton** as **his** secretary of state on Monday. **He** chose **her** because **she** had foreign affairs experience as a **former first lady**.

Barack Obama
his → Refers to **Barack Obama**
He → Refers to **Barack Obama**

Hillary Rodham Clinton
her → Refers to **Hillary Rodham Clinton**
she → Refers to **Hillary Rodham Clinton**
former first lady → Refers to **Hillary Rodham Clinton**

  - **Isaac Newton** and **Albert Einstein** transformed physics. **Newton** developed classical mechanics, while *his* laws of motion shaped early physics. **Einstein** later challenged *his* ideas with relativity, reshaping how **we** understand space and time. **Their** work remains foundational today.

Isaac Newton
Newton → Refers to **Isaac Newton**
his → Refers to **Isaac Newton**
his laws of motion → Refers to **Newton's** contributions to physics
his ideas → Refers to **Newton's** scientific principles

Albert Einstein
Einstein → Refers to **Albert Einstein**
Both Scientists (Newton & Einstein as a Group)
Their work → Refers to **Newton and Einstein's** collective contributions to physics

Coreference and Full Understanding

- **What is Full Text Understanding?**
  - The ability of NLP models to process and extract **meaning** from text. Tasks
  - Involves multiple tasks such as:
    - **Information Extraction –** Identifying key entities, events, and relationships.
    - **Question Answering (QA) –** Responding accurately based on text.
    - **Summarization –** Condensing information while preserving meaning.

- **Example: Coreference Resolution in QA**
  - **Sentence:** *"He was born in 1961."*
  - **Problem:** Who does *"He"* refer to?
  - **Solution:** The model must link *"He"* to a previously mentioned entity (e.g., *Barack Obama*).

Coreference and Full Understanding

- **Gender Differences**
  - Some languages (e.g., **French, Spanish, Arabic**) require **gender agreement** in adjectives and pronouns.
  - Example: *"The doctor is skilled."* → **French:** *"Le médecin est compétent" (M) / "La médecin est compétente" (F).*
- **Number Agreement**
  - Singular/plural forms vary across languages and affect **verbs, adjectives, and pronouns**.
  - Example: *"The children play outside."* → **Spanish:** *"Los niños juegan afuera."*
- **Dropped Pronouns (Pro-Drop Languages)**
  - Some languages **omit pronouns** when they can be inferred from context (e.g., **Chinese, Japanese, Italian**).
  - Example: *"He is going to school."* → **Chinese:** *"去学校了。"* (subject omitted but understood).

Coreference and Full Understanding

- **Gender Differences**
  - Some lar eement in adjective
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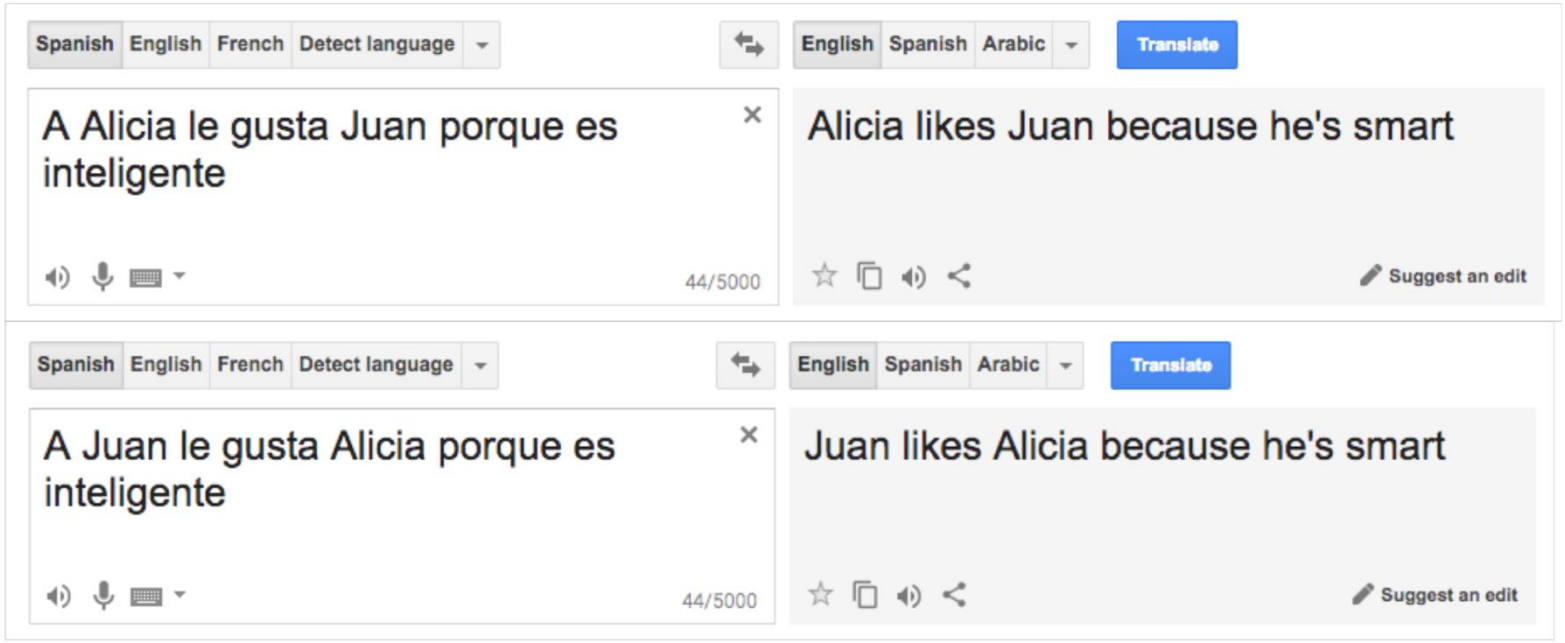

Example: Booking Movie Tickets

- **User Request:** *"Book tickets to see James Bond."*
- **System Response:** *"Spectre is playing near you at 2:00 and 3:00 today. How many tickets would you like?"*
- **User Input (with Errors):** *"Twp toclets fpr the sjpwomg at tjree."*
- **System Tasks:**
  - **Error Handling:** Detect and correct typos → *"Two tickets for the showing at three."*
  - **Intent Recognition:** Understand the **request for two tickets**.
  - **Slot Filling:** Identify the **movie name, time, and number of tickets**.
  - **Confirmation & Booking:** *"You have booked two tickets for Spectre at 3:00 PM."*

Coreference and Full Understanding

- ✅ Enables **better context retention** in NLP applications.
- ✅ Improves accuracy in **machine reading comprehension**.
- ✅ Supports advanced **text-based AI systems** like virtual assistants and chatbots.

Technical Details

Coreference Resolution – Technical Process

- **Step 1: Mention Detection:** Identify **all noun phrases (NPs), pronouns, and named entities** in the text.
  - **Sentence:** *"Marie Curie won the Nobel Prize in Physics. She later won another in Chemistry."*
  - **Detected Mentions:**
    - **Marie Curie** (Proper Noun)
    - **She** (Pronoun)
    - **Nobel Prize in Physics** (Noun Phrase)
    - **another** (Implicit reference)

- **Step 2: Feature Extraction for Mention Pairs:** Extract features for candidate mention pairs to determine coreference. Common Features:
  - **Lexical Similarity** (Exact match or partial match).
  - **Syntactic Agreement** (Gender, number, person).
  - **Semantic Similarity** (Vector-based embedding similarity).
  - **Distance-based Heuristics** (How far apart mentions appear).

Coreference Resolution – Technical Process

- **Step 3: Clustering Coreferent Mentions:** Use **classification models** to predict whether two mentions refer to the same entity. **Approaches:**
  - **Rule-based (Heuristics)** – Hand-crafted rules (e.g., "he" refers to a male entity).
  - **Statistical Methods** – Decision trees, SVMs using syntactic and lexical features.
  - **Neural Networks** – Contextual embeddings (e.g., BERT-based models).

- **Step 4: Coreference Resolution Models**
  - **Mention Pair Models:** Classify whether two mentions are coreferent.
  - **Entity-Level Models:** Cluster all mentions that refer to the same entity.
  - **End-to-End Neural Models:**
    - **BERT-based Models** – Contextual understanding for resolving coreferences.

Example Output with Clusters: *"Marie Curie won the Nobel Prize in Physics. **She** later won **another** in Chemistry."*
Cluster 1: {**Marie Curie, She**} - **Cluster 2:** {**Nobel Prize in Physics, another**}

Types

Mention Pair Models

- A **binary classification approach** to coreference resolution. Determines whether **two mentions refer to the same entity**.
- Each **pair of mentions** in is treated as a **classification instance**.
  1. **Mention Detection** – Identify all noun phrases (NPs) and pronouns.
  2. **Pair Generation** – Form mention pairs (e.g., *"Obama", "He"*).
  3. **Feature Extraction** – Compute features for each mention pair:
     1. **Lexical similarity** (Do words match?)
     2. **Syntactic agreement** (Gender, number match?)
     3. **Semantic similarity** (Are they related in meaning?)
     4. **Distance-based features** (How far apart are the mentions?)
  4. **Binary Classification** – Predict if the two mentions refer to the same entity.
- **Example:** *"Barack Obama gave a speech.* ***He*** *emphasized economic reform."*
  - **Mention Pair:** *("Barack Obama", "He")* → Classified as **coreferent**.

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     4. **Distan**
  4. **Binary Classification** – Predict if the two mentions refer to the same entity.

✅ **Simple and interpretable** model.
✅ **Effective for resolving pronouns and named entities**.
❌ **Computationally expensive** (many mention pairs to classify).
❌ **Ignores global document structure**, leading to inconsistencies.

- **Example:** *"Barack Obama gave a speech. **He** emphasized economic reform."*
  - **Mention Pair:** *("Barack Obama", "He")* → Classified as **coreferent**.

Entity-Level Models

- Instead of classifying individual **mention pairs**, entity-level models group all mentions **into clusters** that refer to the same entity. Focuses on **global consistency** rather than just local pairwise relationships.
  1. **Mention Detection** – Identify noun phrases (NPs) and pronouns in the text.
  2. **Entity Clustering** – Instead of treating mentions as isolated pairs, maintain clusters of mentions referring to the same entity.
  3. **Feature-Based Learning** – Use syntactic, semantic, and discourse-based features to decide which mentions belong to the same cluster.
  4. **Iterative Merging** – Continuously refine entity clusters based on confidence scores.
- **Example:** *"Barack Obama gave a speech.* ***He*** *emphasized economic reform.* ***Obama*** *later met with reporters."*
  - **Entity Cluster:** {Barack Obama, He, Obama}
- **Instead of comparing each pair separately, the model groups all mentions into a single entity representation.**

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- **Ir** **s into a single entity representation.**

✅ **Ensures document-level consistency** in coreference resolution.
✅ **Reduces errors from independent mention pair classifications**.
❌ **More computationally complex** than mention pair models.
❌ **Requires advanced learning techniques** (e.g., entity embeddings, clustering algorithms).

End-to-End Neural Models

- Deep learning-based approaches that **jointly learn** mention detection and coreference resolution.
- Unlike traditional models, they do not rely on **handcrafted features** or **separate mention detection steps**.
  1. **Contextualized Representations** – Uses **pre-trained embeddings** (e.g., BERT, SpanBERT) to encode mentions.
  2. **Span Scoring** – Assigns a probability score to possible mention spans.
  3. **Coreference Scoring** – Computes similarity between mention spans to determine if they belong to the same entity cluster.
  4. **Clustering** – Uses a neural model to **group mentions** into coreference clusters.
- **Example:** *"Marie Curie won the Nobel Prize in Physics.* ***She*** *later won another in Chemistry."*
  - **Model Prediction: C1:** {Marie Curie, She}. C**2:** {Nobel Prize in Physics, another}

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✅ **Fully trainable and data-driven** (no handcrafted rules required).
✅ **Uses deep contextualized embeddings** for better entity understanding.
✅ **End-to-end learning** improves efficiency and reduces error propagation.
⚠️ **Requires large labeled datasets** for effective training.
⚠️ **Computationally expensive** due to deep learning inference costs.

Challenges

When Mention Detection is Not Simple

- **Over-Generating Mentions:** Simply marking all **pronouns, named entities, and noun phrases (NPs)** as mentions can lead to **incorrect extractions**. Some cases do **not** refer to entities in a meaningful way.
  - "**It is funny":** "It" is a dummy pronoun and does not refer to a real entity.
  - "**Every student":** A quantifier phrase, not referring to a specific student.
  - "**No student":** A negative phrase, not pointing to an actual person.
  - "**The best donut in the world":** A superlative phrase that is subjective rather than a concrete entity.
  - "**100 miles":** A numerical phrase, a measurement rather than an entity.

- Mention detection must be **selective and context-aware**.
- Not all **pronouns, named entities, or noun phrases** should be treated as meaningful references.

Handling Bad Mentions

- **Problem: Spurious or Incorrect Mentions**
  - Coreference resolution often detects **mentions that do not actually refer to an entity**.
  - Incorrect mentions introduce **noise** and reduce system accuracy.

- **Approaches to Handle Bad Mentions**

  1. **Train a Classifier to Filter Spurious Mentions**
     - Use **machine learning models** to predict whether a detected mention is valid.
     - Features for classification:
       - **Lexical & syntactic cues** (e.g., named entity recognition, dependency structure).
       - **Contextual embeddings** (e.g., BERT-based features).
       - **Mention frequency & salience** (e.g., is it an important noun phrase?).

Handling Bad Mentions , cont.

- **Approaches to Handle Bad Mentions**
  2. **Candidate Mentions Approach (More Common)**
     - Keep **all detected mentions** as possible candidates.
     - Run the **coreference resolution model** on all candidate mentions.
  3. **Post-Processing: Discard Singleton Mentions:** After the coreference system processes mentions:
     - **Remove singleton mentions** (i.e., mentions not assigned to any coreference cluster).
     - Helps eliminate **irrelevant or one-off references**.

Why This Works?

✅ Avoids introducing errors from **hard filtering upfront**.

✅ **Keeps recall high** while removing **unnecessary mentions post-processing**.

✅ Improves **efficiency** in clustering by discarding **irrelevant data**.

Coreference vs. Anaphora in NLP

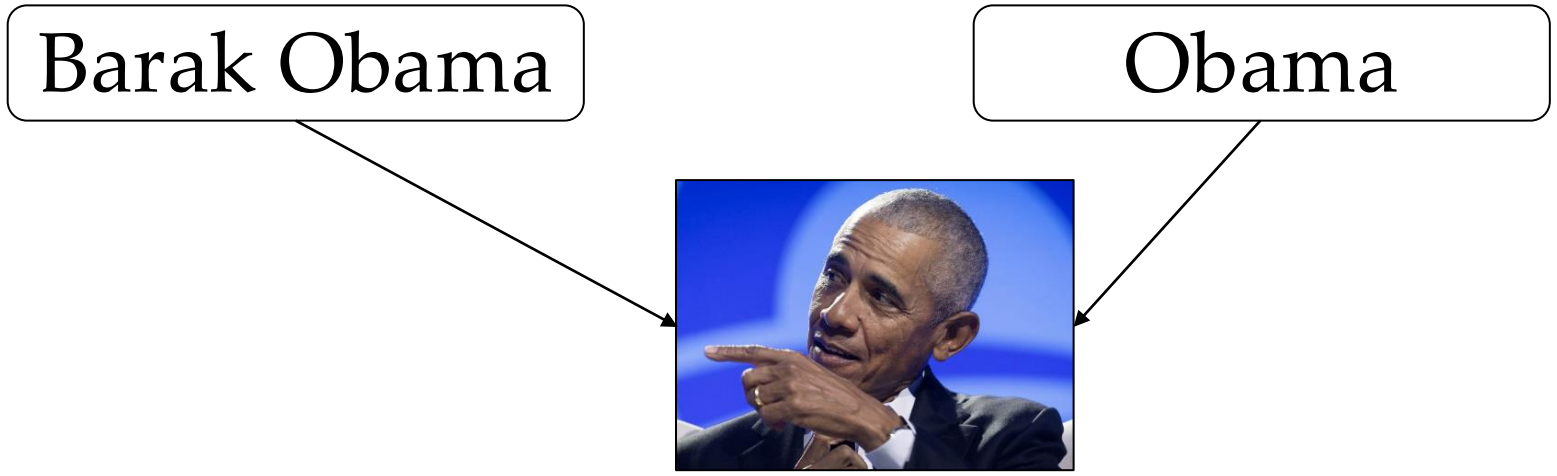
- **Coreference occurs when two mentions refer to the same real-world entity.** Example:
  - *"Barack Obama traveled to Europe.* ***Obama*** *met with world leaders."*
  - **"Obama" refers back to "Barack Obama"**, indicating coreference.
- **Anaphora is a specific type of coreference where an anaphor refers to an antecedent.** The **anaphor's meaning depends on its antecedent**. Example:
  - *"Barack Obama said* ***he*** *would sign the bill."*
  - **"He" (anaphor) refers to "Barack Obama" (antecedent).**
- **Key Differences**
  - **Coreference**: Any two mentions referring to the same entity (proper nouns, pronouns, etc.).
  - **Anaphora**: A term (anaphor) specifically depends on an earlier term (antecedent) for interpretation.

Coreference vs. Anaphora in NLP

- Coreference with named entities

Text

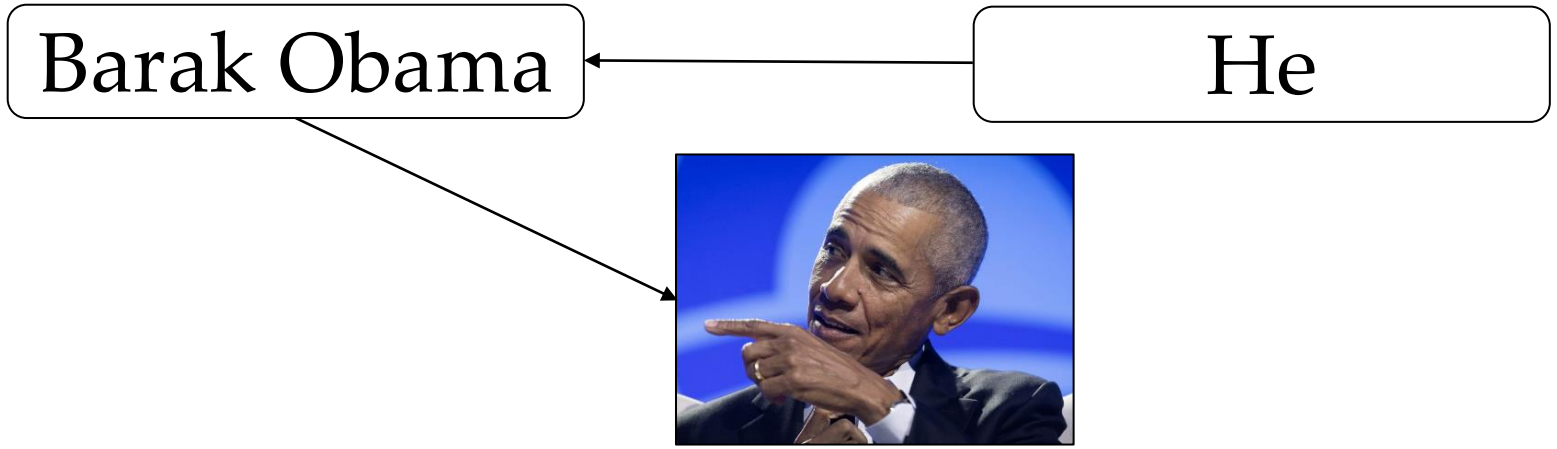
World



- Anaphora

Text

World



Types

Types of Coreference Models

- **Rule-Based Coreference Resolution:** Uses **handcrafted linguistic rules** for **pronominal anaphora resolution**.
  - Example: *"Barack Obama gave a speech. **He** emphasized economic reform."*
    - Rule: "**He**" **refers to the nearest male noun phrase (Barack Obama).**
  - **Pros:** Simple, interpretable. **Cons:** Limited to predefined rules, struggles with complex cases.
- **Mention Ranking Models:** Instead of binary mention pairs, ranks **possible antecedents** for a mention.
  - Example: *"Obama met with leaders. **He** later held a press conference."*
    - Model ranks "**Obama**" as the best antecedent for "**He**".
  - **Pros:** More flexible than mention pair models. **Cons:** Requires additional training for ranking.
- **Mention Pair Models:** Classifies **pairs of mentions** as coreferent or not.
  - Example: Model checks if *("Obama", "he")* belong to the same entity.
  - **Pros:** Simple implementation. **Cons:** Ignores global document structure, leads to inconsistencies.
- **Clustering-Based Models:** Groups **all mentions** referring to the same entity into clusters.
  - Example Output: **Cluster 1**: {Barack Obama, he, his}. **Cluster 2**: {The bill, it}
  - **Pros:** Ensures global consistency. **Cons:** Computationally expensive.

Knowledge-Based Coreference Resolution

- Uses **external knowledge sources** (e.g., **Wikipedia, knowledge graphs**) to resolve ambiguous references.

- Goes beyond **textual context** by incorporating **real-world knowledge**.
  1. **Mention Detection** – Identifies noun phrases and pronouns.
  2. **Entity Linking** – Maps mentions to entities in a **knowledge base** (e.g., Wikidata, DBpedia).
  3. **Coreference Resolution** – Uses structured knowledge to group mentions referring to the same entity.

- **Example:** *"Einstein developed the theory of relativity. The physicist changed our understanding of space-time."*
  - **Without knowledge-based coreference:** "The physicist" could refer to any scientist.
  - **With knowledge-based coreference:** Recognizes "**Einstein**" = "**The physicist**" using knowledge graphs.

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- **Example:** *"Ein st changed our understanding o*
  - **Without know** o any scientist.
  - **With knowle hysicist**" using knowledge gr

✅ **Improves accuracy** by resolving ambiguous references.
✅ **Handles named entities effectively**.
⚠️ **Requires large-scale knowledge bases**.
⚠️ **Computationally expensive** for real-time applications.

Comparison

Coreference Model	Description	Pros	Cons
Rule-Based (Pronominal Anaphora Resolution)	Uses predefined linguistic rules to resolve pronouns.	Simple, interpretable	Limited to predefined rules, struggles with complex cases
Mention Pair	Classifies pairs of mentions as coreferent or not.	Simple implementation	Ignores global document structure, leads to inconsistencies
Mention Ranking	Ranks possible antecedents for a mention instead of binary pairs.	More flexible than mention pair models	Requires additional training for ranking
Clustering	Groups all mentions referring to the same entity into clusters.	Ensures global consistency	Computationally expensive
Knowledge-Based Coreference	Uses external knowledge sources (e.g., Wikipedia, knowledge graphs) to resolve coreferences.	Improves accuracy with world knowledge, resolves ambiguous cases	Requires large-scale knowledge bases, high computational cost
Entity-Level Models	Groups all mentions into entity clusters instead of classifying mention pairs.	Ensures document-level consistency, reduces errors	More computationally complex than mention pair models
End-to-End Neural Models	Uses deep learning to jointly detect and resolve coreferences.	Fully trainable, uses contextual embeddings, more accurate	Requires large labeled datasets, high computational cost

Example

Mention Pair Running Example

- Example: "I voted for Nader because he was most aligned with my values," he said.

I | Nader | he | my | she

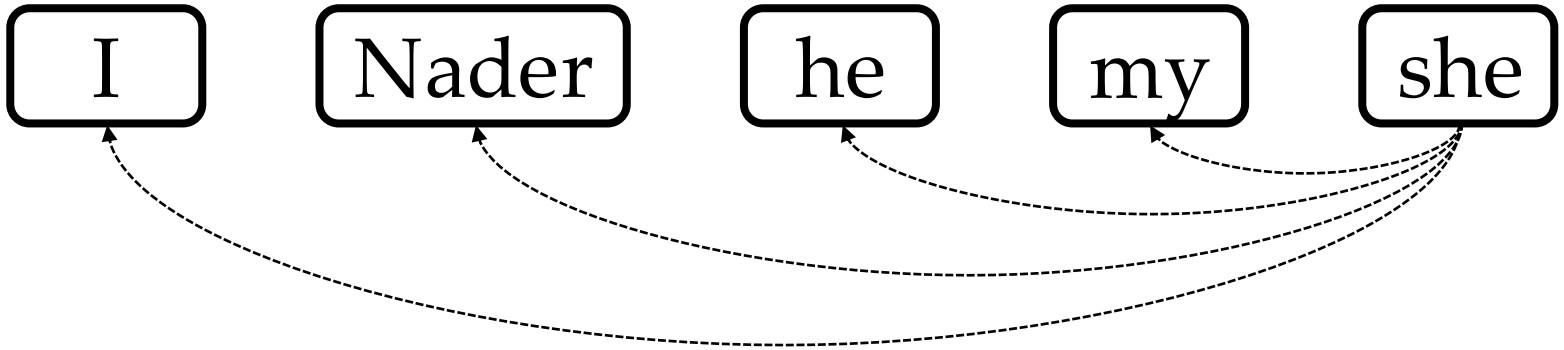
Coreference Cluster 1

Coreference Cluster 2

Mention Pair Running Example

- Train a binary classifier that assigns every pair of mentions a probability of being coreference: $p(m_i, m_j)$
  - Example: for "she" look at all candidate antecedents (previous mentions) and decide which are coreference with it

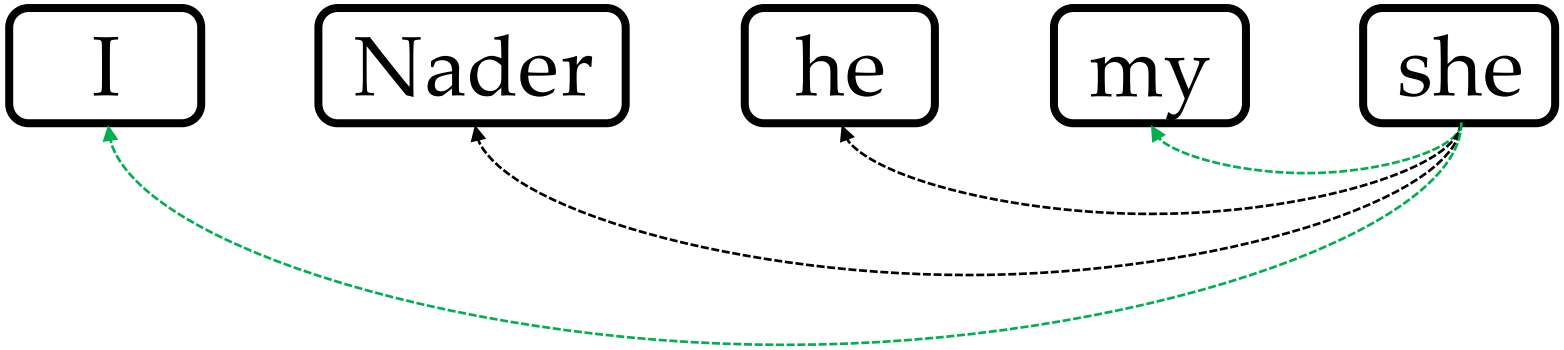
"**I** voted for **Nader** because **he** was most aligned with **my** values," **she** said.



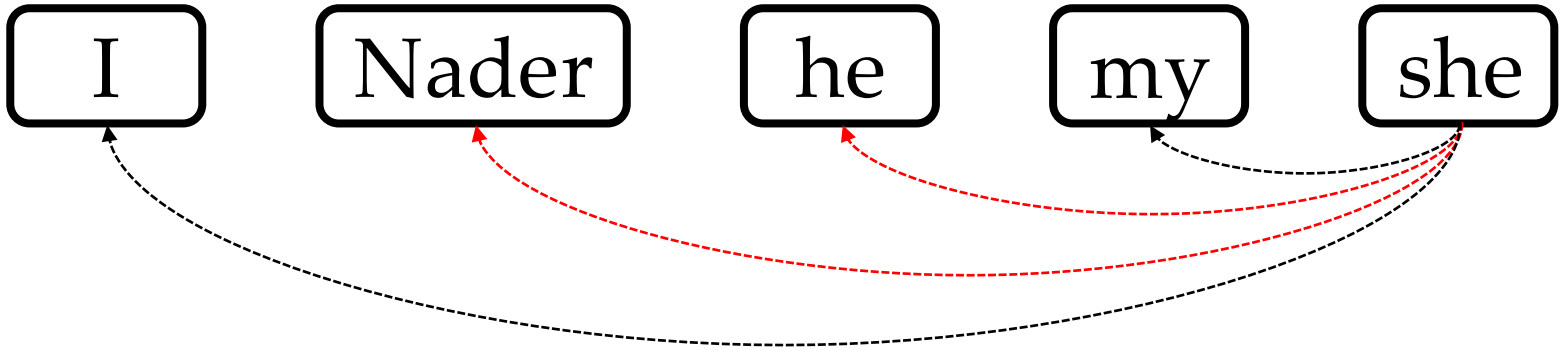
Coreference with she?

Mention Pair Running Example

Positive examples: want $p(m_i,m_j)$ to be near 1



"**I** voted for **Nader** because **he** was most aligned with **my** values," **she** said.



Negative examples: want $p(m_i,m_j)$ to be near 0

Mention Pair Running Example

- N mentions in a document
- y_{ij} = 1 if mentions m_i and m_j are coreferent, -1 if otherwise
- Just train a regular cross-entropy loss (looks a bit different because it is binary classification)

$$J = -\sum_{i=2}^{N} \sum_{j=1}^{i} y_{ij} \log p(m_j, m_i)$$

Iterate through mentions

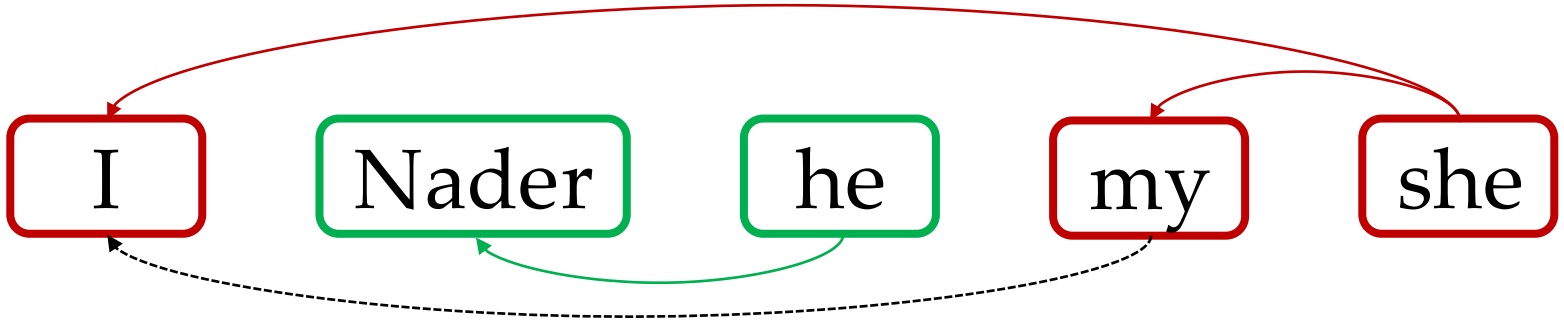
Iterate through candidate antecedents (previously occurring mentions)

Coreferent mentions pairs should get high probability, others should get low probability

Mention Pair Running Example

- Coreference resolution is a clustering task, but we are only scoring pairs of mentions. What to do?
  - Pick some threshold (e.g., 0.5) and add coreference links between mention pairs where $p(m_i, m_j) >$ threshold
  - Take the transitive closure to get the clustering

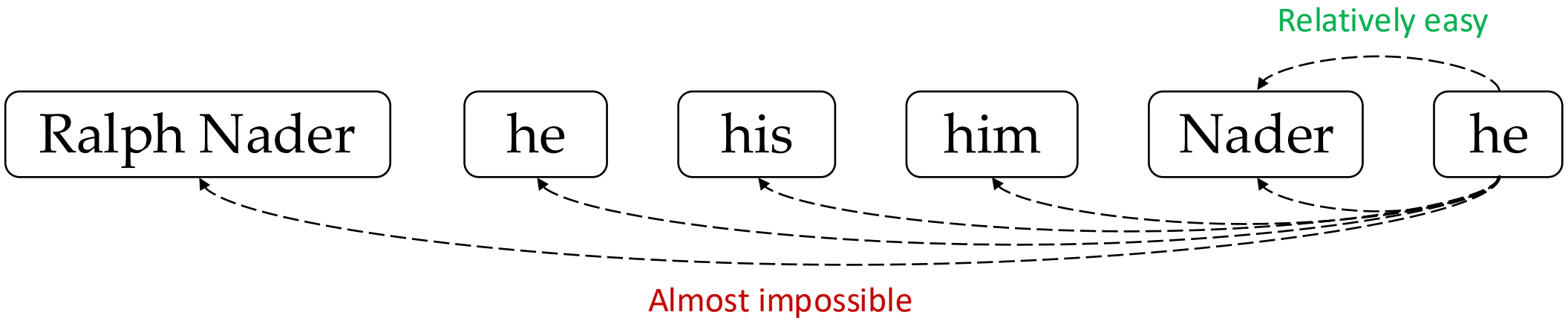
"**I** voted for **Nader** because **he** was most aligned with **my** values," **she** said.



Even though the model did not predict this coreference link, I and my are coreference due to transitivity.

Disadvantages

- Suppose we have a long document with the following mentions
  - Ralph Nader … he … his … him … <several paragraph> .. voted for Nader because he



- Many mentions only have one clear antecedent
  - But we are asking the model to predict all of them
- Solution: train a model to predict only one antecedent for each mention

Mention Ranking

- Assign each mention its **highest scoring** candidate antecedent according to the model
- Dummy **NA** mention allows model to decline linking the current mention to anything (**singleton** or **first mention**)

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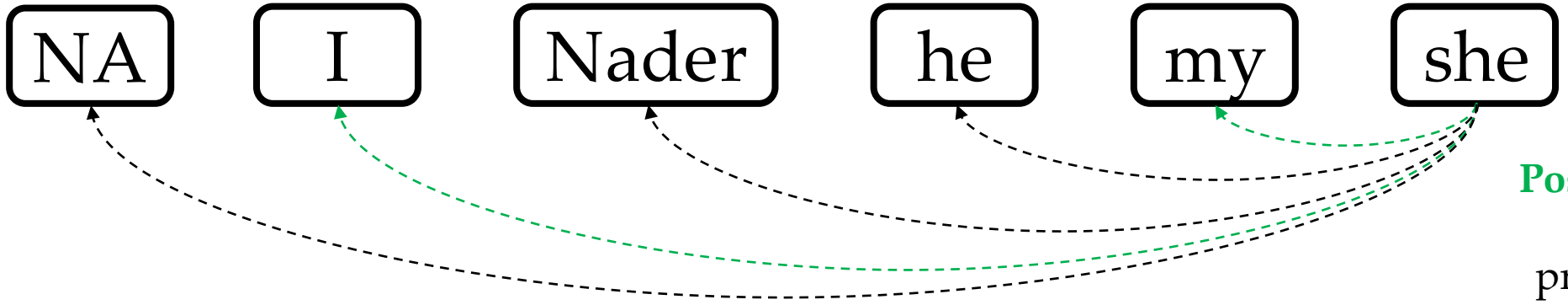

Positive examples: model has to assign a high probability to either one (but not necessarily both)

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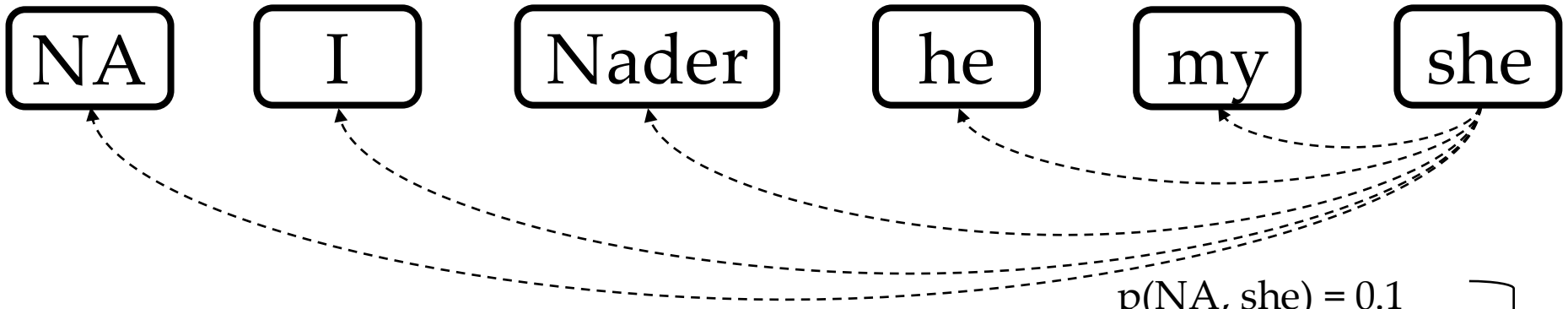


Best antecedents for **she**?

p(NA, she) = 0.1
p(I, she) = 0.5
p(Nader, she) = 0.1
p(he, she) = 0.1
p(my, she) = 0.2

Apply softmax over the scores for candidate antecedents so probabilities sum up to 1.

Calculating Probability

- **Non-Neural Statistical Classifier**
  - Uses **hand-crafted features** and **logistic regression** or **SVMs** to assign scores.
  - Computes **probability** based on feature weights:

$$P(a|m) = \frac{e^{W \cdot F(a,m)}}{\sum_{a'} e^{W \cdot F(a',m)}}$$

  - Where:
    - *a* is the antecedent
    - *m* is the mention
    - *W* is the feature weight ve
    - *F*(*a*,*m*) are the extracted features

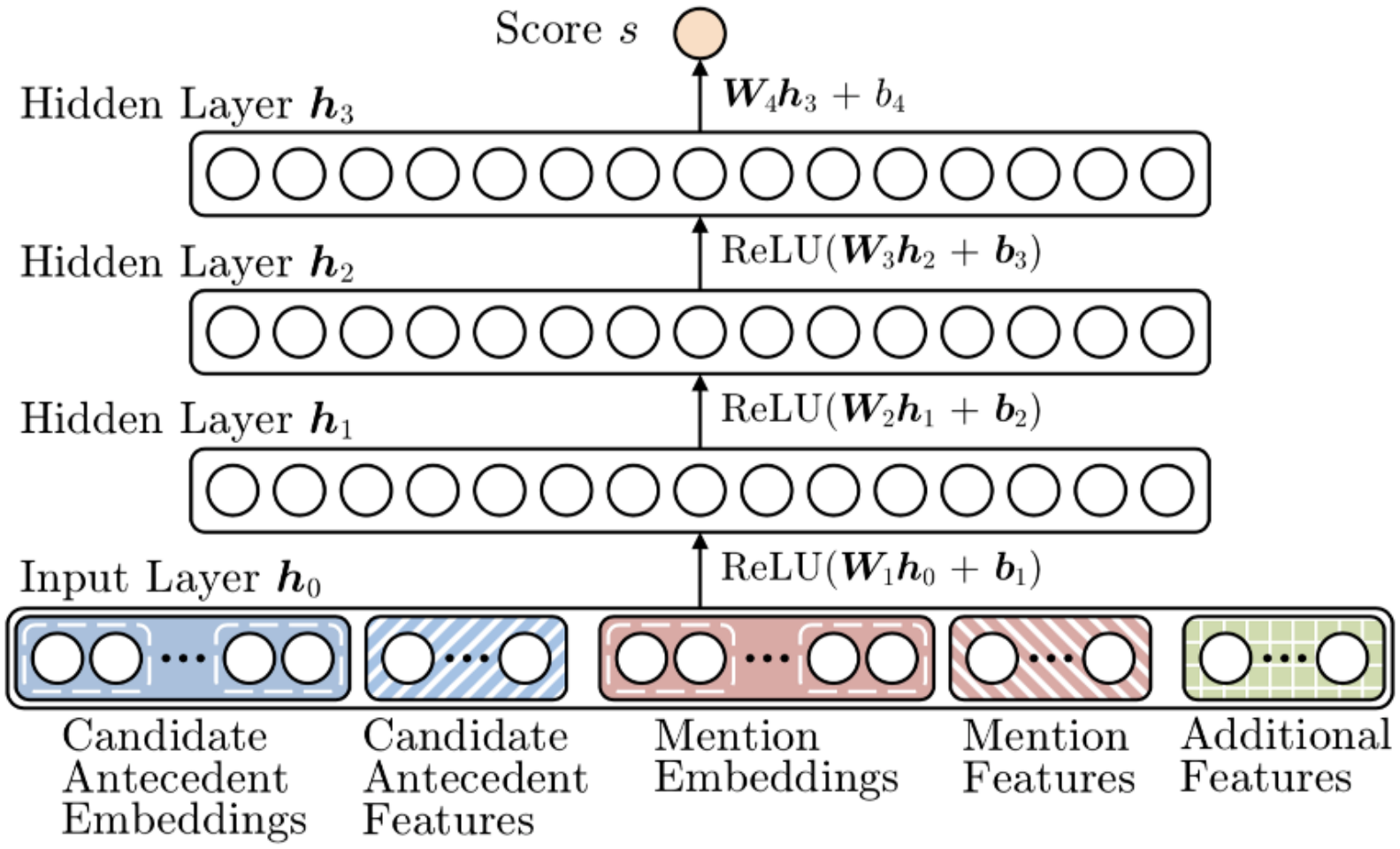
Calculating Probability

- **Simple Neural Network**
  - F**eedforward neural network** to compute mention-antecedent score.
  - Given a mention *m* and antecedent *a,* the probability is computed as:

$$P(a|m) = \text{softmax}(f(W \cdot [m, a]))$$

  - Where:
    - *f* is a non-linear activation function (e.g., ReLU)
    - *W* is a learnable weight matrix

Calculating Probability (Neural Network)



Calculating Probability

- **Advanced Model with LSTMs & Attention**
- Uses **bidirectional LSTMs (BiLSTMs)** to encode mention and antecedent context.
- Applies **attention mechanisms** to capture long-range dependencies.
- The probability is calculated as:

$$P(a|m) = \text{softmax}(v^T \cdot \tanh(W_1 h_m + W_2 h_a + W_3 \cdot \text{attention}))$$

- Where:
  - h_m and h_a are contextualized representations from BiLSTMs
  - Attention weights help focus on important antecedents

Non-Neural Model Features

- Person/number/gender agreement
  - Jack gave **Mary** a gift. **She** was excited.
- Semantic compatibility
  - **.. The mining conglomerate… the company**
- Certain syntactic constraints
  - John bought **him** a new car. [*him cannot be John*]
- More recently mentioned entities preferred for reference
  - **Jon** went to a movie. **Jack** went as well. **He** was not busy.
- Grammatical role: prefer entities in the subject position
  - **John** went to a movie with **Jack**. **He** was not busy.
- Parallelism: **John** went with **Jack** to movie. **Joe** went with **him** to a bar.

Neural Model Input

- Embeddings: previous two words, first word, last word, head word, .. of each mention.
  - The **head** word is the "most important" word in the mention – you can find it using a parser; e.g., The fluffy cat stuck in the tree.

- Still need some other features
  - Distance
  - Document genre
  - Speaker information

Takeaways

Takeaways

- Coreference links mentions to enable **coherent document-level understanding**.
- Pairwise models are simple but ignore global consistency.
- Ranking and clustering models improve **document-level coherence**.
- End-to-end neural models leverage **contextual embeddings (e.g., BERT)**.
- Effective coreference requires **agreement features, discourse cues, and sometimes external knowledge**.